



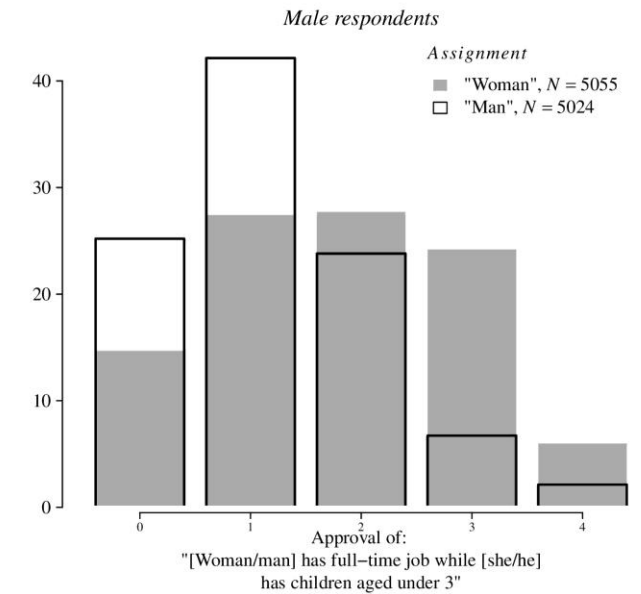
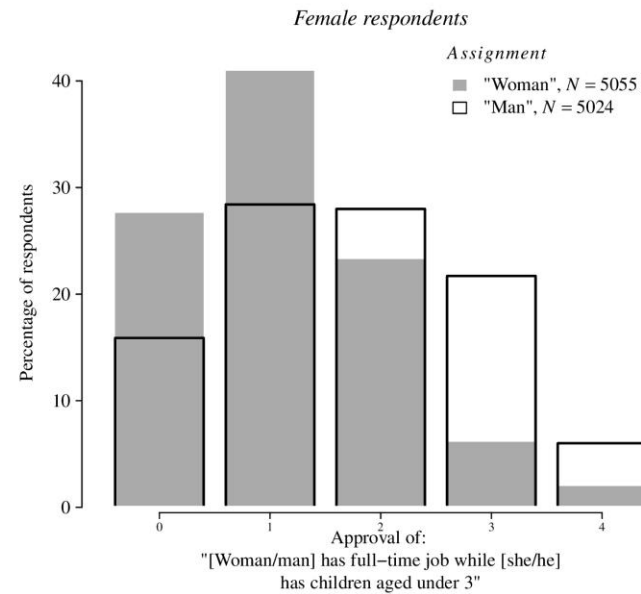
Replication Project:
***“Household education gaps
and gender role attitudes”***
Giani, Hope & Skorge (2022)

Original Paper - Introduction

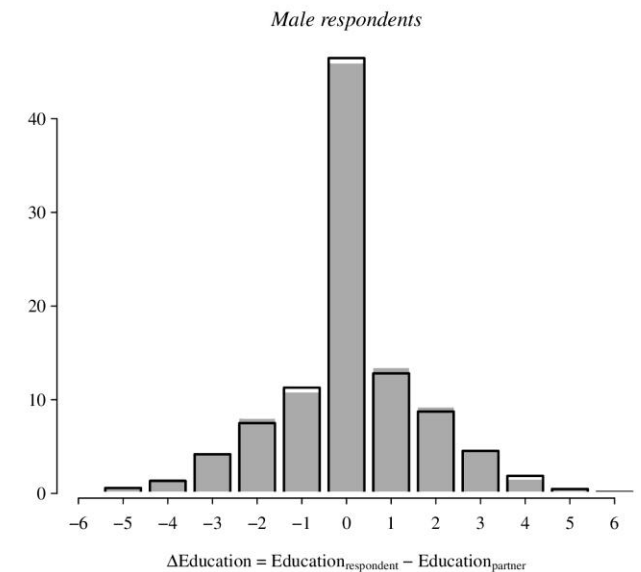
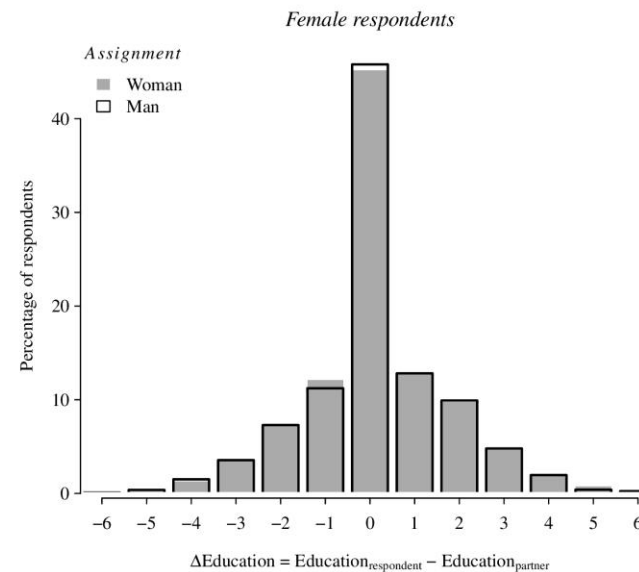
- It's a 2022 paper, published in the *Political Science Research and Methods* by Giani, Hope & Skorge.
- It investigates how educational gap within the household mediate one's gender childcare bias, hereby expressed as the difference in approval of a mother versus a father having a full-time job while having children under the age of three.
- The findings align with a resource-bargaining framework.

Data – European Social Survey

- 2018 ESS (Round 9) data, face-to-face survey carried out in 19 EU countries.
- Dependent variable of interest: Approval. On a scale from 0 (Strongly approve) to 4 (Strongly Disapprove).
- Main independent variables:
 - *Vignette Gender*: half of the respondents were asked about a mother and half about a father, binary variable.
 - *Household Education Gap*: the difference in the respondent education minus that of their partner, discrete variable spanning from -6 to +6.



Note: Approval ranges from Approval (1) to Disapproval (5).



Note: Positive values indicate respondents are more educated than their partner.

Original Model: Gaussian Kernel Interflex (Hainmueller, Mummolo and Xu, 2019)

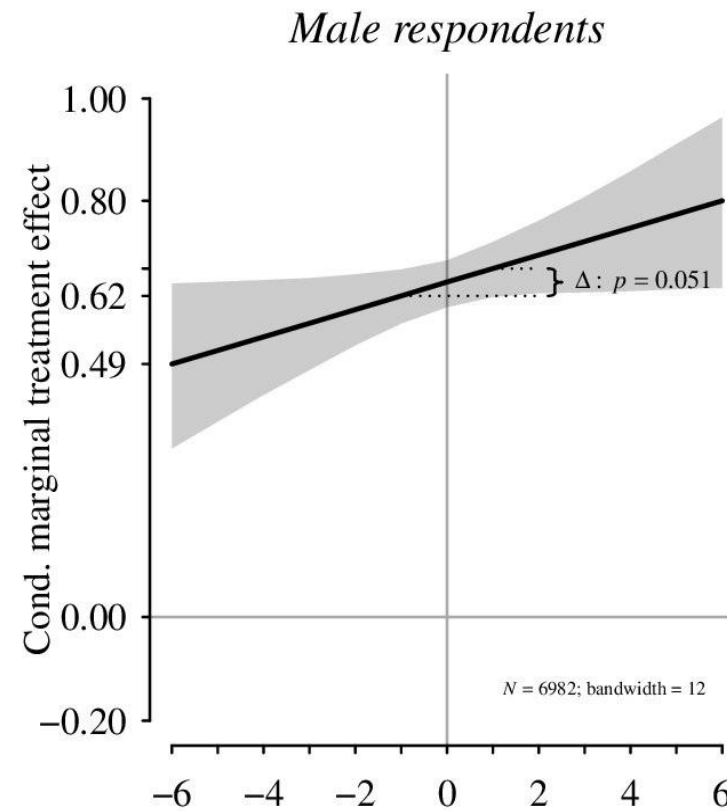
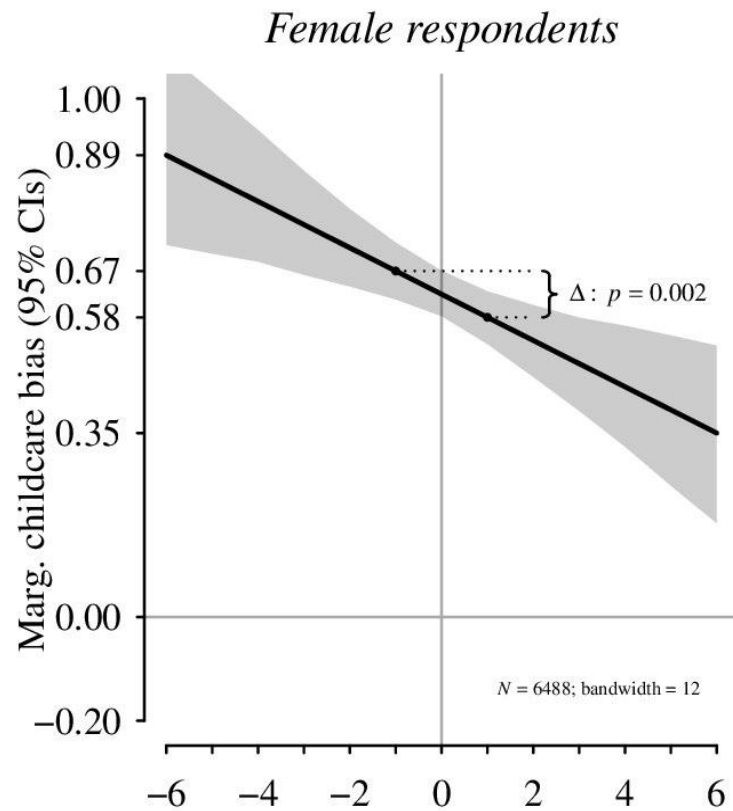
The model allows the gender childcare bias to vary non-monotonically across levels of the household educational gap:

$$+ \gamma(\Delta E\{i, c\}) Z\{i, c\} + \mu c + \varepsilon i, c\} Y\{i, c\} \\ = \alpha + f(\Delta E\{i, c\}) + g(\Delta E\{i, c\}) B\{i, c\}$$

The key quantity of interest is $g(\cdot)$: the marginal gender childcare bias, the difference in approval between a mother and a father holding a full-time job, conditional on $\Delta E_{i,c}$.

The sample is split by respondent gender to assess whether the pattern differs between men and women.

```
### Analysis with covariates for women --> core
ix.kern.z.w <- interflex(dt.w,
  Y="Approval",
  D="Vignette_gender",
  X="Edu_diff",
  Z=v.z,
  estimator="kernel",
  FE=c("Country"),
  parallel=v.parallel,
  cores=v.cores,
  nboots=v.boots,
  neval=13,
  cutoffs = c(-6,-1,.9,6),
  na.rm=TRUE,
  wald=TRUE,
  diff.values=c(-1,1),
  bw = 12
)
## For men --> core
ix.kern.z.m <- interflex(dt.m,
  Y="Approval",
  D="Vignette_gender",
  X="Edu_diff",
  Z=v.z,
  estimator="kernel",
  FE=c("Country"),
  parallel=v.parallel,
  cores=v.cores,
  nboots=v.boots,
  neval=13,
  cutoffs = c(-6,-1,.9,6),
  na.rm=TRUE,
  wald=TRUE,
  diff.values=c(-1,1),
  bw = 12
)
```



Main Findings

1. For *female respondents*, as the household education gap increase in their favour, their gender childcare bias tends to diminish.
2. For *male respondents* the opposite trend is observed: as the educational gap increases in their favour, they tend to have a bigger gender bias.
3. These findings are in line with a resource-bargaining framework: women who out-educate their partners hold more power in the household and hold more egalitarian attitudes; men show the opposite pattern.

My Contribution: Linear Interflex Model

```
## Females --> Linear
ix.z.w.lin <- interflex(dt.w,
  Y="Approval",
  D="Vignette_gender",
  X="Edu_diff",
  FE=c("Country"),
  Z=v.z,
  estimator="linear",
  parallel=v.parallel,
  cores=v.cores,
  nboots=v.boots,
  diff.values=c(-1,1),
  na.rm = TRUE
)
```

```
## Males --> Linear
ix.z.m.lin <- interflex(dt.m,
  Y="Approval",
  D="Vignette_gender",
  X="Edu_diff",
  Z=v.z,
  estimator="linear",
  FE=c("Country"),
  parallel=v.parallel,
  cores=v.cores,
  nboots=v.boots,
  diff.values=c(-1,1),
  na.rm=TRUE,
)
```

- The Gaussian kernel model is computationally intensive and its output, a smooth function $g(\cdot)$, is not straightforward to interpret.
- Given that the original results appeared linear, I fit a linear interflex model retaining the same covariates and country fixed effects, trading flexibility for parsimony and interpretability.

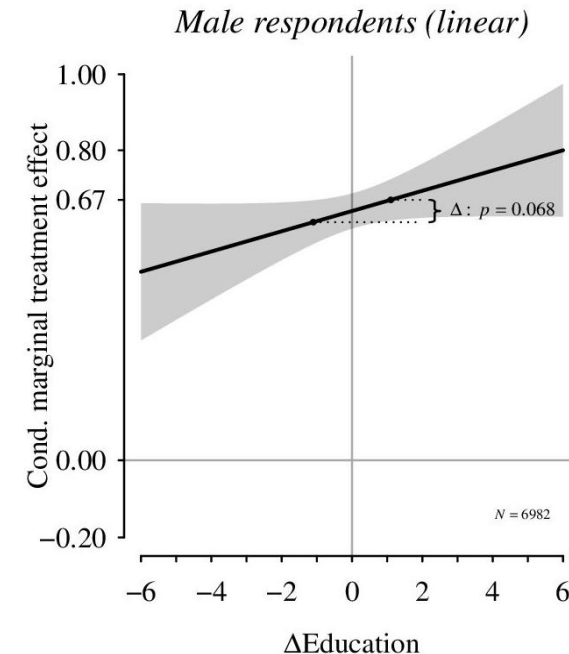
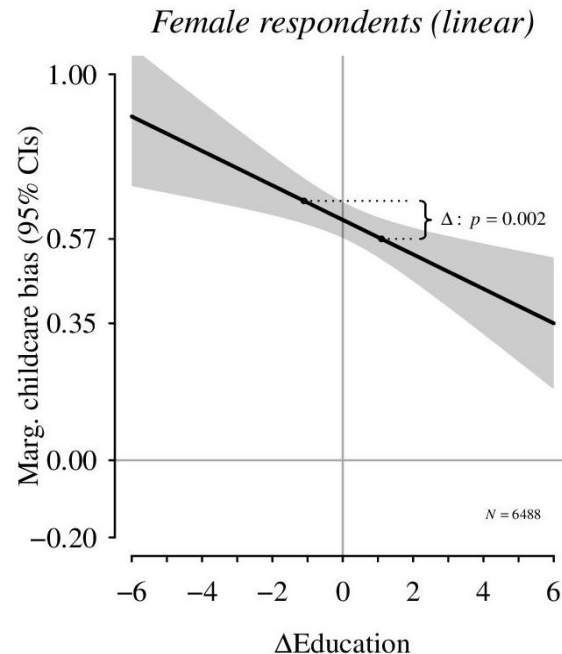
$$Y\{i, c\} = \alpha + \beta_1 B\{i, c\} + \beta_2 \Delta E\{i, c\} + \beta_3 (B\{i, c\} \Delta E\{i, c\}) + \gamma Z\{i, c\} + \mu c + \varepsilon\{i, c\}$$

In this specification our focus is β_3 , the marginal gender childcare bias given a specific level of household educational gap.

Table 1: Linear Interflex Models - Core Explanatory Variables

	<i>Dependent variable:</i>	
	Approval	
	Women (Linear)	Men (Linear)
	(1)	(2)
Educational Gap (β_1)	0.035*** (0.011)	0.013 (0.011)
Vignette Gender (β_2)	0.623*** (0.024)	0.646*** (0.023)
Edu. Gap $\times$ Vignette Gender (β_3)	-0.045*** (0.015)	0.026* (0.014)
Observations	6,488	6,982
R ²	0.258	0.245
Adjusted R ²	0.254	0.241
Residual Std. Error	0.943 (df = 6450)	0.961 (df = 6944)

Note: *p<0.1; **p<0.05; ***p<0.01



My Findings

- The two models reports identical results for female respondents and strongly similar values for males
- Specifically, we find that has a β_3 values of -0,045 for women and 0,026 for men; the base value, when the educational gap is equal to 0, is similar for both genders, validating the resource-bargaining framework .

In conclusion: The results are near-identical across both specifications, suggesting the added complexity of the kernel model was unnecessary.



Q & A

Thank you for your attention